

## CASSIOPEIA

## Cassiopeia



SIZE RANKING 25

BRIGHTEST STARS  
Schedar ( $\alpha$ ) 2.2,  
Gamma ( $\gamma$ ) 2.2

GENITIVE Cassiopeiae

ABBREVIATION Cas

HIGHEST IN SKY AT 10 PM  
October–DecemberFULLY VISIBLE  
90°N–12°S

This distinctive constellation of the northern sky is found within the Milky Way between Perseus and Cepheus and north of Andromeda. The large W shape formed by its five main stars is easily recognizable. It is an ancient Greek constellation, representing the mythical Queen Cassiopeia of Ethiopia.

## SPECIFIC FEATURES

Gamma ( $\gamma$ ) Cassiopeiae (see p.285) is a hot, rapidly rotating star that occasionally throws off rings of gas from its equator, which causes unpredictable changes in its brightness. It has ranged between magnitudes 3.0 and 1.6, but it currently lies at magnitude 2.2, which makes it the joint brightest star in the constellation.

A variable with a more predictable cycle is Rho ( $\rho$ ) Cassiopeiae, an intensely luminous, yellow-white supergiant that fluctuates between 4th and 6th magnitudes every 10 or 11 months. It is estimated that it lies more than 10,000 light-years away, which is exceptionally distant for a naked-eye star.



Eta ( $\eta$ ) Cassiopeiae is an attractive stellar pair consisting of a yellow and a red star. Its components are of magnitudes 3.5 and 7.5 and can be seen through a small telescope. This pair forms a true binary; the fainter companion orbits the brighter star every 480 years.

Cassiopeia contains a number of open clusters within range of small instruments. Chief among them is M52 (see p.290), near the border with Cepheus. It is visible through binoculars as a somewhat elongated patch of light, and its individual stars—including a bright orange giant at one edge—can be seen through a small telescope. M103 is a small, elongated group best viewed through a small telescope. Nearby is a larger cluster, NGC 663, which is more suitable for binocular observation. NGC 457 is a looser star cluster containing the 5th-magnitude star Phi ( $\phi$ ) Cassiopeiae. This cluster's appearance has been likened to an owl—its two brightest stars mark the owl's eyes.

## M103

M103's main feature is a chain of three stars like a mini Orion's belt. The northernmost member of the line (top) is not a true member of the cluster but lies closer to Earth.



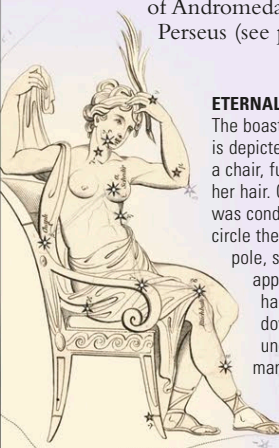
## M52

Through binoculars, this cluster appears as a misty patch about one-third the diameter of a full moon. A telescope is needed to resolve its individual stars.

## MYTHS AND STORIES

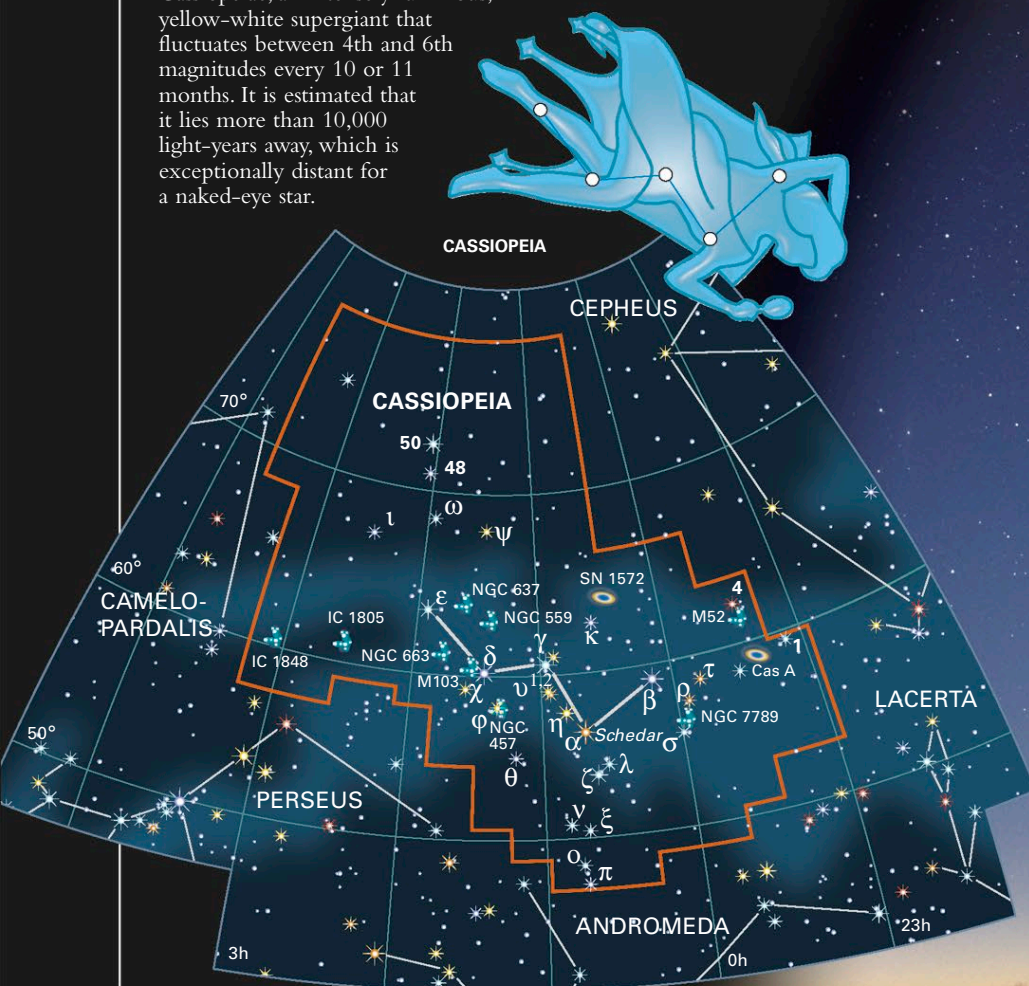
## THE VAIN QUEEN

Wife of Cepheus and mother of Andromeda, Queen Cassiopeia was notoriously vain. She enraged the Nereids, daughters of Poseidon, by boasting she was more beautiful. In punishment, Poseidon sent a sea monster to ravage her kingdom, which eventually led to the rescue of Andromeda by Perseus (see p.368).



## ETERNAL VANITY

The boastful queen is depicted sitting in a chair, fussing with her hair. Cassiopeia was condemned to circle the celestial pole, sometimes appearing to hang upside down in an undignified manner.



## POLAR POINTER

The distinctive W shape formed by the main stars of Cassiopeia is easy to locate in the sky. The center of the W points toward the north celestial pole.